Prof. Stefan Minner
Principal Editor, *International Journal of Production Economics*

Resubmission of a substantially revised manuscript (previously PROECO-D-26-05160)
Replenishing perishables on shared delivery routes: Stock-age-based inventory policies, stochastic lookahead and decision-focused learning

Dear Professor Minner,

Thank you for your decision on PROECO-D-26-05160. You noted that the manuscript did not take a clear perishable inventory management perspective and therefore fell short in comparisons with state-of-the-art methods. I agree with both points. Rather than revising the text, I rebuilt the study around them; the new manuscript shares only the problem domain with the rejected version. I submit it as a new manuscript and summarise the changes below.

1. Perishable inventory management perspective.

- The problem is now a periodic-review perishable inventory system with a routing coupling: every day each policy observes the age-structured stock of every store, decides replenishment, and ships it on shared, capacitated routes; demand is served FIFO, unmet demand is lost and stock past its shelf life is discarded. The rejected version fixed a multi-day plan without observing stock, which is not how perishables are managed.

- Results are reported in inventory terms—cost composition, fill rate, waste rate, delivery frequency and store-level service—and analysed across three network layouts, shelf life, lost-sales/waste cost ratio, fleet tightness, routing cost, demand variability, issuing policy (FIFO vs. partial LIFO) and a case with observed fresh-food demand (FreshRetailNet-50K), with 100 test episodes per setting, three training seeds and prespecified, Holm-adjusted hypotheses.

- The structural results are stated in inventory terms and corrected: the recourse admits FIFO-optimal solutions once early disposal is allowed (the original lemma was false without it), the marginal value of a delivery is bounded by the lost-sales and waste penalties, and the decision-focused fixed point reduces to the perishable newsvendor fractile. All results are verified numerically.

2. Comparison with state-of-the-art methods.

- The comparator set now contains the perishable inventory rules recommended in the literature—periodic base-stock and base-stock with estimated waste (Broekmeulen and van Donselaar, 2009; Haijema and Minner, 2019)—the deliver-up-to-level rule of Crama et al. (2018) in its state-dependent form, a budgeted robust lookahead (Bertsimas and Thiele, 2006; Solyalı et al., 2012), and the Scenario Predict-then-Optimize method of Jia et al. (2025), together with mean and tuned-quantile lookahead and a perfect-information reference. All policies use the same information and the same forecasts and are tuned on separate validation data.

- Every lookahead decision is obtained from an exact MILP solved to a proven gap of at most 0.5%; the rejected version relied on heuristic fallbacks.

- The proposed method is now decision-focused learning applied on top of the strongest inventory rule: UL-DFL learns context- and stock-age-dependent order-up-to levels for the deliver-up-to-level rule from the realised cost of whole episodes. Decision-focused learning on top of the deterministic lookahead (LA-DFL) is retained and accompanied by an ablation with day-specific quantile levels.

- Across the settings, decision-focused learning applied on top of the deliver-up-to-level rule (UL-DFL) is the cheapest policy in eleven of thirteen settings (tied with its rule-free ablation in a twelfth; scenario-based lookahead is cheaper only when routing is expensive), and cheaper than the rule it starts from in every setting, significantly in eleven; the learned corrections are interpretable (lower levels on promotion days and after the weekly demand peak). Decision-focused learning applied to the deterministic lookahead improves it, but its gain is reproduced by a day-specific quantile level, which we report as an ablation.

Scope and transparency. The synthetic demand is generated by a documented contextual model; the real-data case uses public data of one retailer, and its preparation (stock-out correction, rescaling to the network's capacities, exclusion of products with strong level shifts) is described in full. Earlier claims about M5 calibration, an 81-cell factorial, regret against a heuristic clairvoyant plan and a solver speed-up have been withdrawn. Code, seeds and result files reproducing every number are provided.

I hope that the revised study now meets the scope and standards of IJPE. It has not been published and is not under consideration elsewhere.

Sincerely,

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